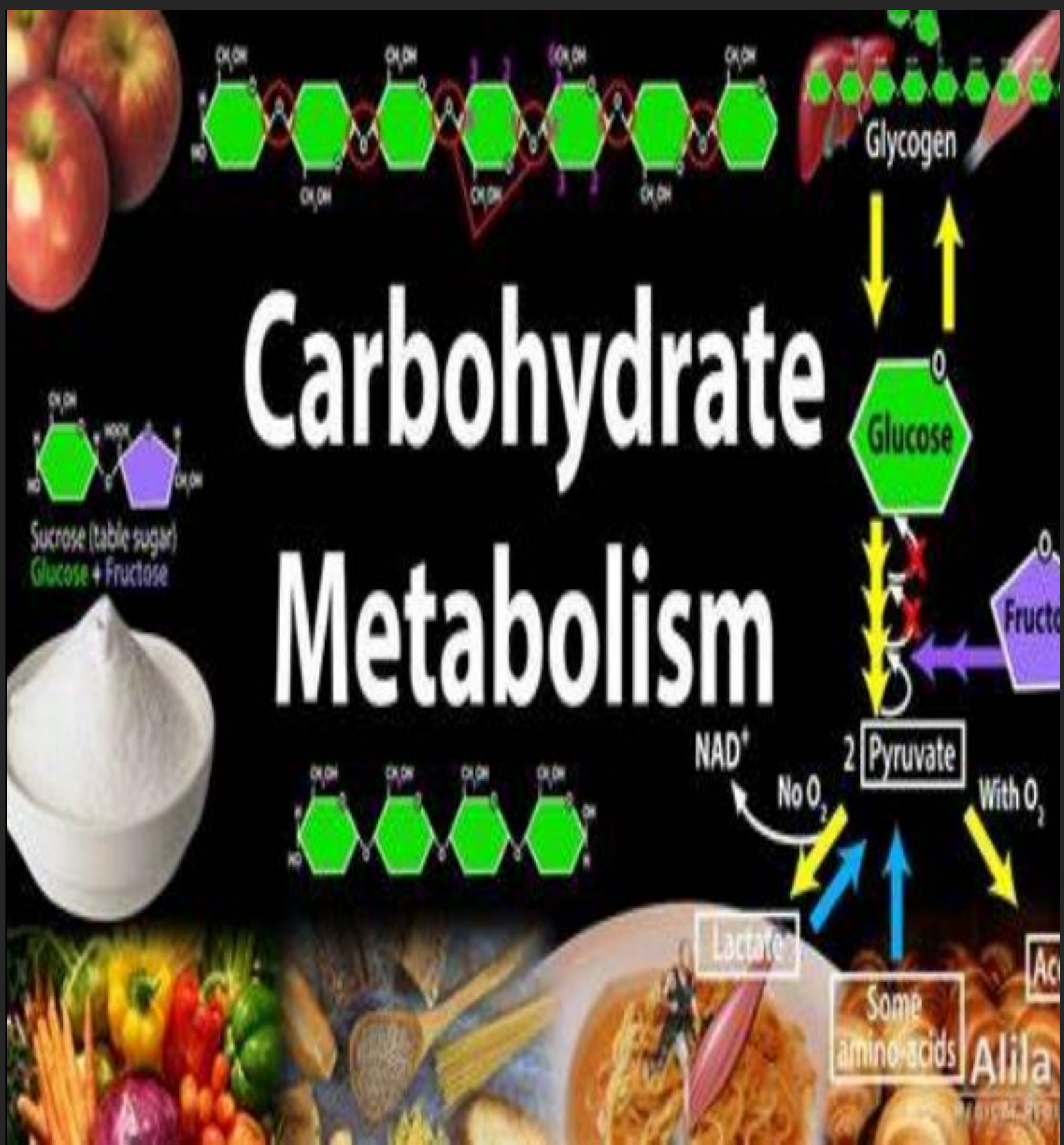




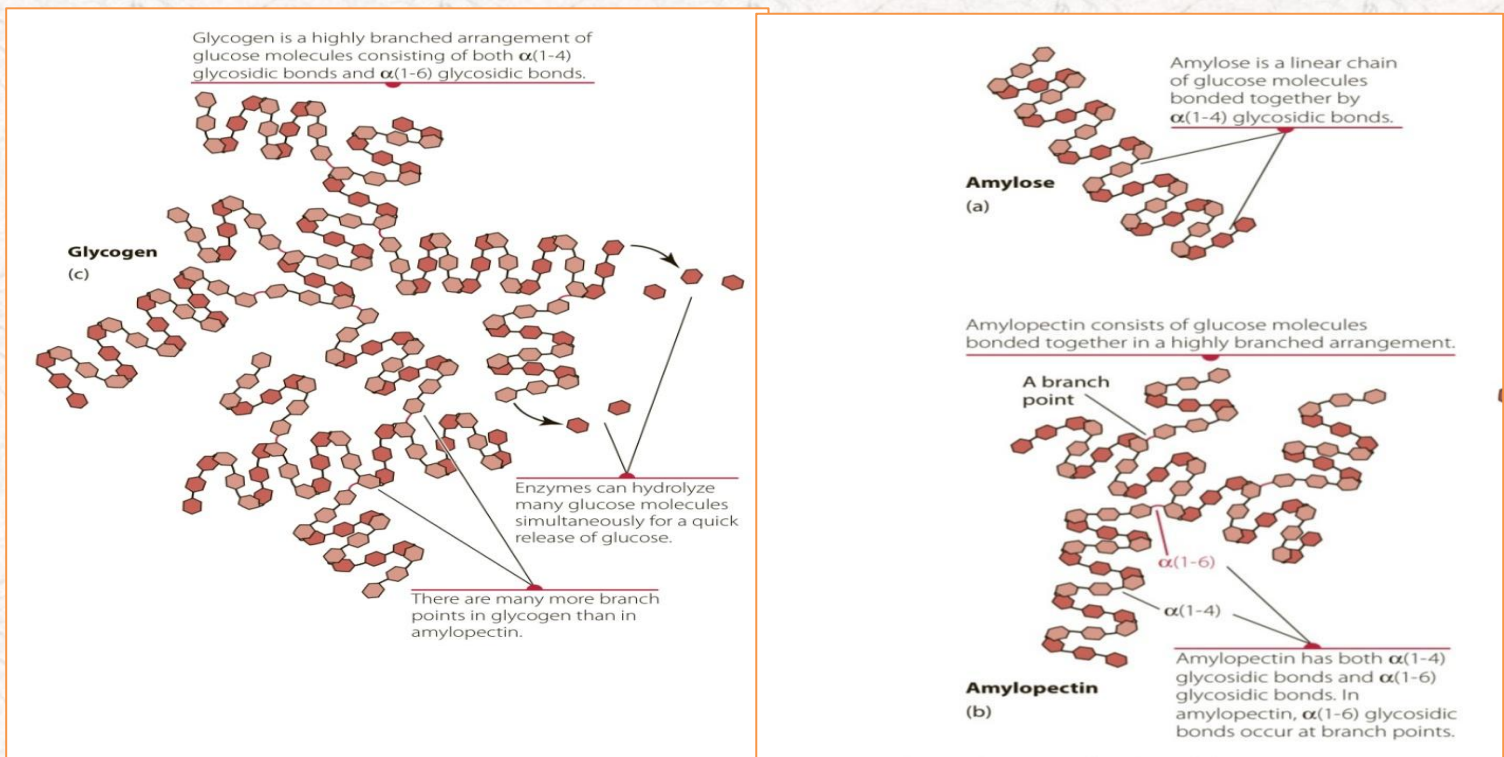
Carbohydrate Metabolism

Outline

- ©Carbohydrate digestion
- ©Carbohydrate absorption
- ©Carbohydrate post absorption transport
- ©Carbohydrate metabolism

©Carbohydrate digestion

- The major source of energy in the average human diet is carbohydrate, supplying half or more of the total caloric intake.
- Roughly half of dietary carbohydrate is in the form of polysaccharides such as starches and dextrins, derived largely from cereal grains and vegetables.
- The remaining half is supplied as simple sugars, the most abundant being sucrose, followed by lactose, maltose, glucose, and fructose.
- Polysaccharides and disaccharides are the most abundant carbohydrates in the food supply, although some free glucose and fructose are present in honey, certain fruits and vegetables, "invert" sugar used in confections, and highfructose corn syrup.
- Before dietary carbohydrates can be used by the body's cells, they must first be absorbed from the gastrointestinal (GI) tract into the bloodstream, a process normally restricted to monosaccharides—the form of carbohydrates enterocytes can absorb.
- Poly-, tri-, and disaccharides therefore must be hydrolyzed.
- The hydrolytic enzymes involved are collectively called **glycosidases** or, alternatively, **carbohydrases**.



Structure of starch and glycogen

© Digestion of Polysaccharides

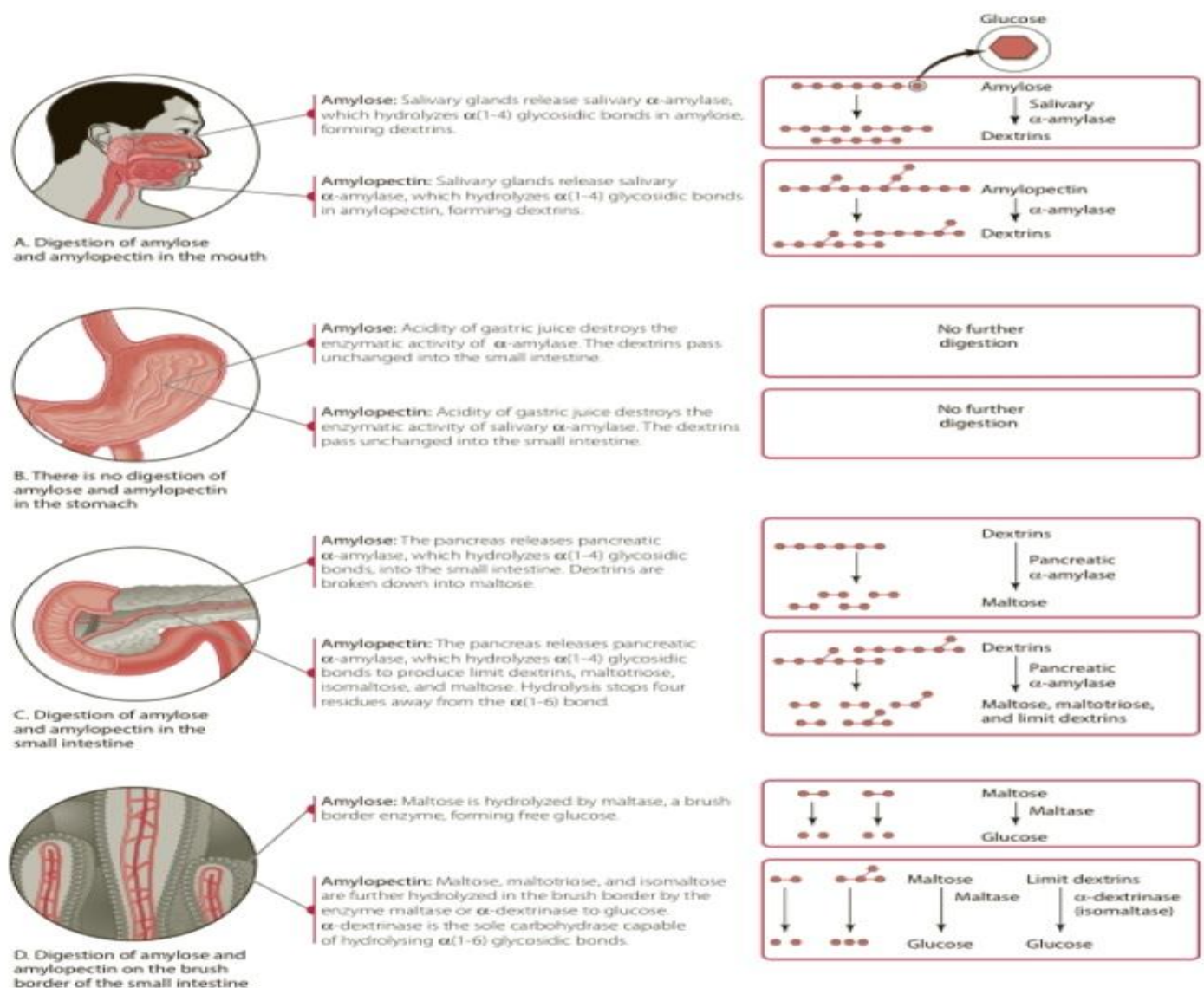
- The digestion of polysaccharides (starches) starts in the mouth. The key enzyme is salivary α -amylase, a glycosidase that specifically hydrolyzes $\alpha(1-4)$ glycosidic linkages.
- The $\beta(1-4)$ bonds of cellulose, the $\beta(1-4)$ bonds of lactose, and the $\alpha(1-6)$ linkages that form branch points in the starch amylopectin are resistant to this enzyme.
- Given the short period of time that food is in the mouth before being swallowed, this phase of digestion produces few mono- or disaccharides.
- However, the salivary amylase action continues in the stomach until the gastric acid penetrates the food bolus and lowers the pH sufficiently to inactivate the enzyme.
- The starches move into the duodenum and jejunum, where they are acted upon by pancreatic α -amylase.

- The presence of pancreatic bicarbonate in the duodenum elevates the pH to a level favorable for enzymatic function.
- The α -amylase hydrolyzes $\alpha(1-4)$ glycosidic bonds in both amylose and amylopectin to produce oligosaccharides (also called dextrins or limit dextrins), maltose, and maltotriose
- The branched oligosaccharides, trisaccharide maltotriose, and maltose are further digested by specific enzymes in the brush border. α -amylase can further break down the oligosaccharides to maltose and maltotriose.
- The partially hydrolyzed amylopectin is not fully digested by α -amylase; this enzyme's action stops several residues short of the $\alpha(1-6)$ bonds, leaving a limit dextrin.
- These limit dextrins are acted on by α -dextrinase (also called isomaltase), which is attached to the brush border membrane.
- This enzyme contains two polypeptides with two active sites, one with specificity to $\alpha(1-4)$ linkages and the other with specificity to both $\alpha(1-4)$ and $\alpha(1-6)$ linkages.
- α -dextrinase is the only intestinal enzyme that will hydrolyze $\alpha(1-6)$ glycosidic bonds.
- Glucose is released from limit dextrins by the combined action of α -dextrinase and other brush border enzymes.
- A portion of the starch of beans and certain vegetables and other resistant starches are not fully digested.
- This is partially due to the accessibility of the food to the enzyme and partially related to naturally occurring amylase inhibitors in some foods.
- α -amylase inhibitors can be used to impede digestion of dietary starch and thus reduce nutrient absorption as a means to combat the overweight and obesity problem

©Digestion of Disaccharides

- Virtually no digestion of disaccharides or small oligosaccharides occurs in the mouth, stomach, or lumen of the small intestine.
- Digestion takes place almost entirely within the microvilli (the brush border) of the upper small intestine via disaccharidase activity, and the resulting monosaccharides immediately enter the enterocytes with the facilitation of specific transporters .
- Among the enzymes located on the enterocytes are lactase, sucrase, maltase, and trehalase.
- Lactase catalyzes the cleavage of lactose to equimolar amounts of galactose and glucose.
- Lactose has a $\beta(1-4)$ linkage, and lactase is stereospecific for this β linkage.
- Lactase activity is high in infants, but in most mammals, including humans, it decreases a few years after weaning.
- This diminishing activity can lead to lactose malabsorption and intolerance.
- The frequency of lactose intolerance in human populations varies widely depending on geography, race, and ethnicity.
- The highest frequency is seen in Native Americans and in people of Asian, African, and Middle Eastern descent.
- The lowest frequency is seen in white individuals originating from northern European countries.
- Many lactose-free products are available for individuals with lactose intolerance.

- Additionally, lactase can be added directly to regular milk products to hydrolyze the lactose. Sucrase hydrolyzes sucrose to yield one glucose and one fructose residue.
 - Maltase hydrolyzes maltose to yield two glucose units.
 - Trehalase is a brush border disaccharidase that hydrolyzes the $\alpha(1-1)$ glycosidic bonds of trehalose to yield two molecules of glucose.
- **In summary**, nearly all dietary starches and disaccharides ultimately are hydrolyzed completely by specific glycosidases to their constituent monosaccharide units. Monosaccharides, together with small amounts of remaining disaccharides, can then be absorbed by the absorptive cells of the intestine.



©Carbohydrate absorption

Intestinal Absorption of Glucose and Galactose

- After carbohydrate digestion, glucose and galactose are absorbed into the enterocyte by the same mechanisms involving both active transport and facilitated transport.
- The relative contribution of active transport versus facilitated transport depends on the amount of carbohydrate consumed; facilitated transport participates to a greater extent following a large carbohydrate meal.

Active Transport

- The active transport mechanism for glucose and galactose absorption into enterocytes requires energy as ATP and the involvement of a specific transporter protein, designated **sodium-glucose transporter 1 (SGLT1)**.
- The SGLT1 is positioned on the intestinal lumen side of the enterocyte (in the brush border membrane) and simultaneously transports one molecule of glucose (or galactose) and two molecules of Na^+ in the same direction and is thus a **symporter**.
- A mutation in the *SGLT1* gene is associated with glucose-galactose malabsorption.
- The SGLT1 protein has two binding sites: One binds Na^+ and the other binds glucose. The glucose binding site is not available unless the transport protein has already bound Na^+ .
- The attachment of Na^+ to the carrier increases the transport protein's affinity for glucose.
- Sodium is moving down a concentration gradient because the intracellular concentration of Na^+ is low

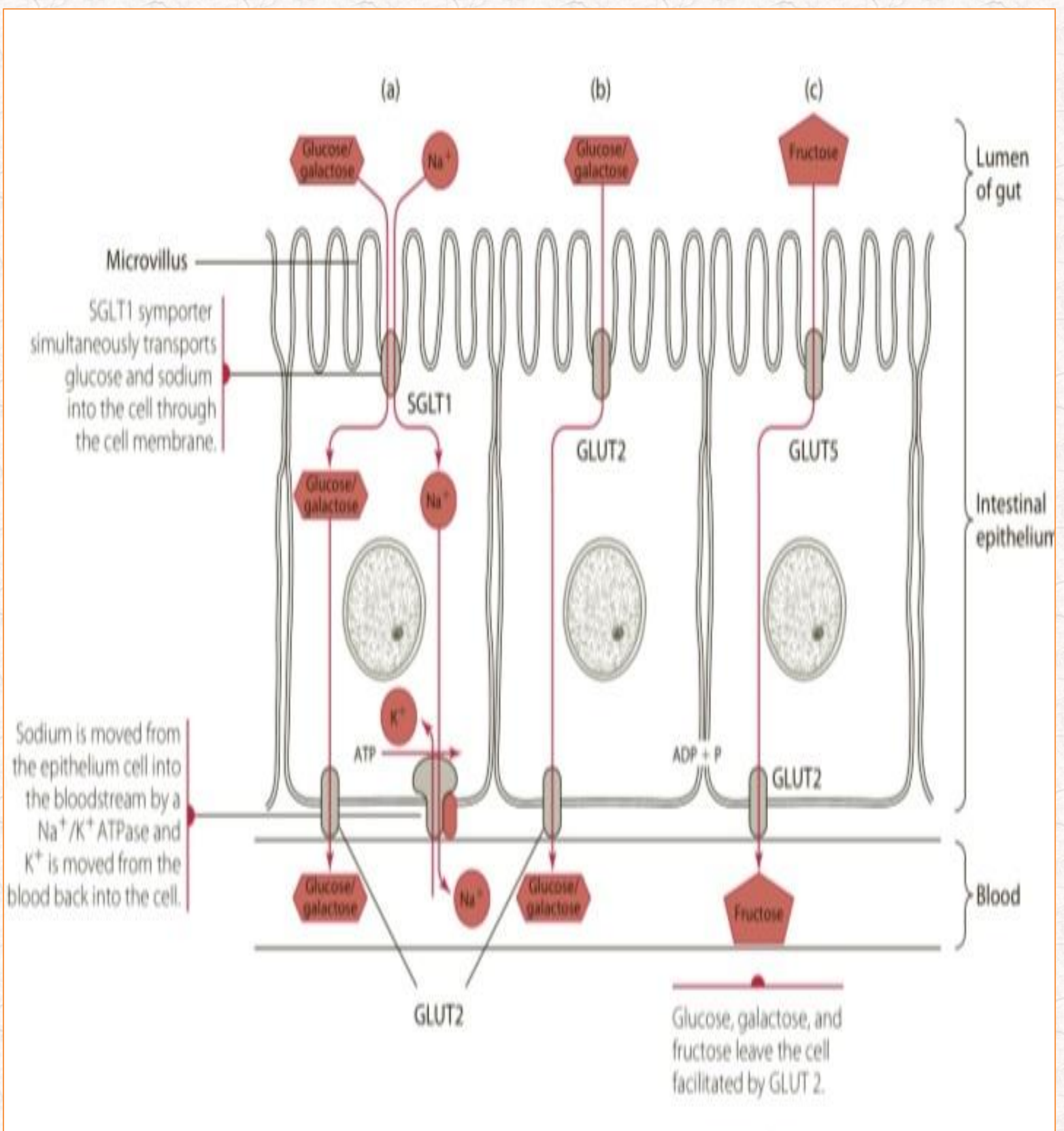
- After Na^+ and glucose are transported into the enterocyte, they are released from SGLT1.
- As the intracellular concentration increases, glucose binds to another transporter in the basolateral membrane, designated glucose transporter type 2 (GLUT2).
- GLUT2 is a low-affinity high-capacity transporter that facilitates the exit of glucose and other hexoses from the enterocyte into the underlying capillaries for delivery into the hepatic portal vein.

Facilitated Transport

- Some glucose (and galactose) can be absorbed into the enterocyte independent of SGLT1 and thus without the input of energy.
- When glucose concentration in the intestinal mucosa is high, such as after the ingestion of a large carbohydrate-containing meal, glucose is transported into the enterocyte by GLUT2 in the brush border membrane.
- When large amounts of glucose enter the enterocyte, intracellular GLUT2 is translocated to the brush border membrane by the movement of the cytoskeleton and the contraction of myosin.
- After high-carbohydrate meals, more glucose is transported into the enterocyte by facilitated transport than by active transport via SGLT1.

Intestinal Absorption of Fructose

- The primary mechanism for fructose transport into the enterocyte is via a specific facilitative transporter, **GLUT5**
- GLUT5 has a high affinity for fructose and is not influenced by the presence of glucose
- Fructose absorption does not require energy (i.e., Na^+ -ATP-dependent transport by SGLT1).
- The rate of uptake of fructose is much slower than that of both glucose and galactose.
- As the intracellular concentration increases, fructose is transported from the enterocyte into the hepatic portal vein by GLUT2 in the basolateral membrane, the same transporter that moves glucose out of the cell.
- The facilitative transport process can proceed only down a concentration gradient.
- At typical dietary intakes, there is no fructose in the systemic circulation due to efficient removal by the liver, where it is phosphorylated and trapped in the hepatocytes.
- Although fructose is absorbed more slowly than glucose or galactose, which are actively absorbed, it is absorbed faster than sugar alcohols such as sorbitol and xylitol, which are absorbed purely by passive diffusion.



Carbohydrate absorption

Post-Absorption Facilitated Transport

- Following the intestinal absorption of glucose, galactose, and fructose, they enter the hepatic portal vein, where they are carried directly to the liver.
- Essentially all of the galactose and fructose is taken up by the liver through specific facilitated transporters and metabolized, whereas only 30-40% of glucose is taken up by the liver, with the majority passing through into the systemic circulation.
- This explains why glucose, but not galactose or fructose, is found in the peripheral blood and why the latter sugars are not directly subject to the strict hormonal regulation that is such an important part of glucose homeostasis.
- Galactose is largely converted glucose derivatives and stored as liver glycogen .
- In contrast, the majority of fructose enters an alternative pathway and is catabolized for energy according to the liver's energy demand.
- If an occasional meal is high in fructose, and the liver's energy needs have been met, excess fructose is converted to triacylglycerol and transported out of the liver for distribution to muscle and adipose tissue.
- However, diets chronically high in fructose can cause **hyperlipidemia** and triacylglycerol accumulation in the liver.
- Glucose is nutritionally the most abundant monosaccharide because it is the exclusive constituent of starch and also occurs in each of three major disaccharides
- The portion of dietary glucose taken up by the liver can be used for energy, stored as glycogen, or returned to the blood during nonfed periods.
- The remainder of the glucose passes into the systemic blood supply and is then distributed among other tissues, such as muscle, kidneys, brain, and adipose tissue.

Glucose Transporters

- Glucose is effectively used by a wide variety of cell types under normal conditions, and its concentration in the blood must be precisely controlled.
- Glucose plays a central role in metabolism and cellular homeostasis.
- Most cells in the body are dependent upon a continuous supply of glucose to supply energy in the form of ATP.
- The signs and symptoms associated with diabetes mellitus are a graphic example of the consequences of a disturbance in glucose homeostasis.
- The cellular uptake of glucose requires that it cross the plasma membrane of the cell.
- The highly polar glucose molecule cannot move across the cellular membrane by simple diffusion because it cannot pass through the nonpolar matrix of the lipid bilayer.
- For glucose to be used by cells, an efficient transport system for moving the molecule into and out of cells is essential.
- In certain cells, such as epithelial cells of the small intestine and renal tubule, glucose crosses the plasma membrane against a concentration gradient by active transport, pumped by an Na^+K^+ -ATPase symport system (SGLT1).
- However, not all cells require an energydependent transporter for glucose uptake.
- A family of protein carriers functions in the facilitated transport of glucose and other monosaccharides, and is called glucose transporters, abbreviated **GLUT**.
- A total of 14 glucose transport proteins have been identified, along with the genes that code for them.

GLUT1 :was the first GLUT identified and the most intensely studied. As the most ubiquitously expressed GLUT, GLUT1 is responsible for the basic supply of glucose to erythrocytes, endothelial cells of the brain, and most fetal tissue. It supplies the glucose to the developing central nervous system during embryogenesis.

GLUT2: is a low-affinity, high-capacity transporter with predominant expression in the β -cells of the pancreas, liver, small intestine, and kidney. As discussed previously, GLUT2 is involved in the transport of glucose and fructose from enterocytes into the portal blood, and when the concentration of glucose in the intestinal lumen is high, it transports glucose and fructose into the enterocyte. The rate of transport is highly dependent upon the blood glucose concentration. In the pancreas, GLUT2 appears to be the sensitive indicator of blood glucose levels and is involved in the release of insulin from the β -cells. High insulin levels cause GLUT2 to leave the plasma membrane of the enterocyte and return to storage vesicles.

GLUT3: is a high-affinity glucose transporter with predominant expression in those tissues that are highly dependent upon glucose, such as the brain and neurons. It is also expressed in cells and tissues that have a high requirement for glucose such as spermatozoa, the placenta, and preimplantation embryos. Some data suggest that a possible dysregulation of GLUT3 might lead to glucose deficits in the brain and thus to dyslexia in children.

GLUT4 :is the primary means by which insulin regulates the cellular uptake of glucose in muscle and adipose tissue. Other cells and tissues such as the liver, kidneys, erythrocytes, and brain do not express GLUT4 and therefore are not dependent upon insulin for glucose uptake. One of the actions of insulin is to cause the translocation of GLUT4 from GLUT4 storage vesicles

GLUT5 :is specific for the transport of fructose and will not transport glucose. It is expressed primarily in the small intestine, but to a lesser degree in kidney, brain, skeletal muscle, and adipose tissue also.

GLUT6: (which was formerly designated GLUT9) is expressed primarily in the brain, spleen, and peripheral leukocytes. It appears to transport hexoses only at higher concentrations.

GLUT7 :has been found in the small intestine and colon. It has a high affinity for glucose and fructose but does not bind galactose or xylose.

GLUT8: (formerly GLUTXI) is expressed mainly in the testis with lower levels in the brain, adrenal gland, liver, spleen, brown adipose tissue, and lung. This GLUT has been studied mostly in animal models.

GLUT9: is primarily detected in the liver and kidney, with lower levels found in the small intestine, placenta, lung, and leukocytes. The metabolic importance of GLUT9 is not fully understood. In addition to a high affinity for glucose and fructose, GLUT9's primary function appears to be the transport of uric acid.

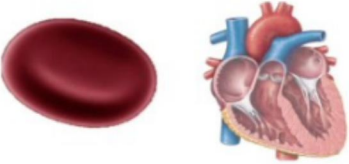
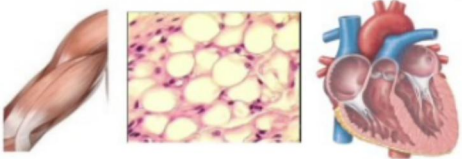
GLUT10 :is present in the heart, lung, brain, liver, skeletal muscle, pancreas, placenta, and kidney. Its physiological role in humans is under investigation and is not fully understood at this time.

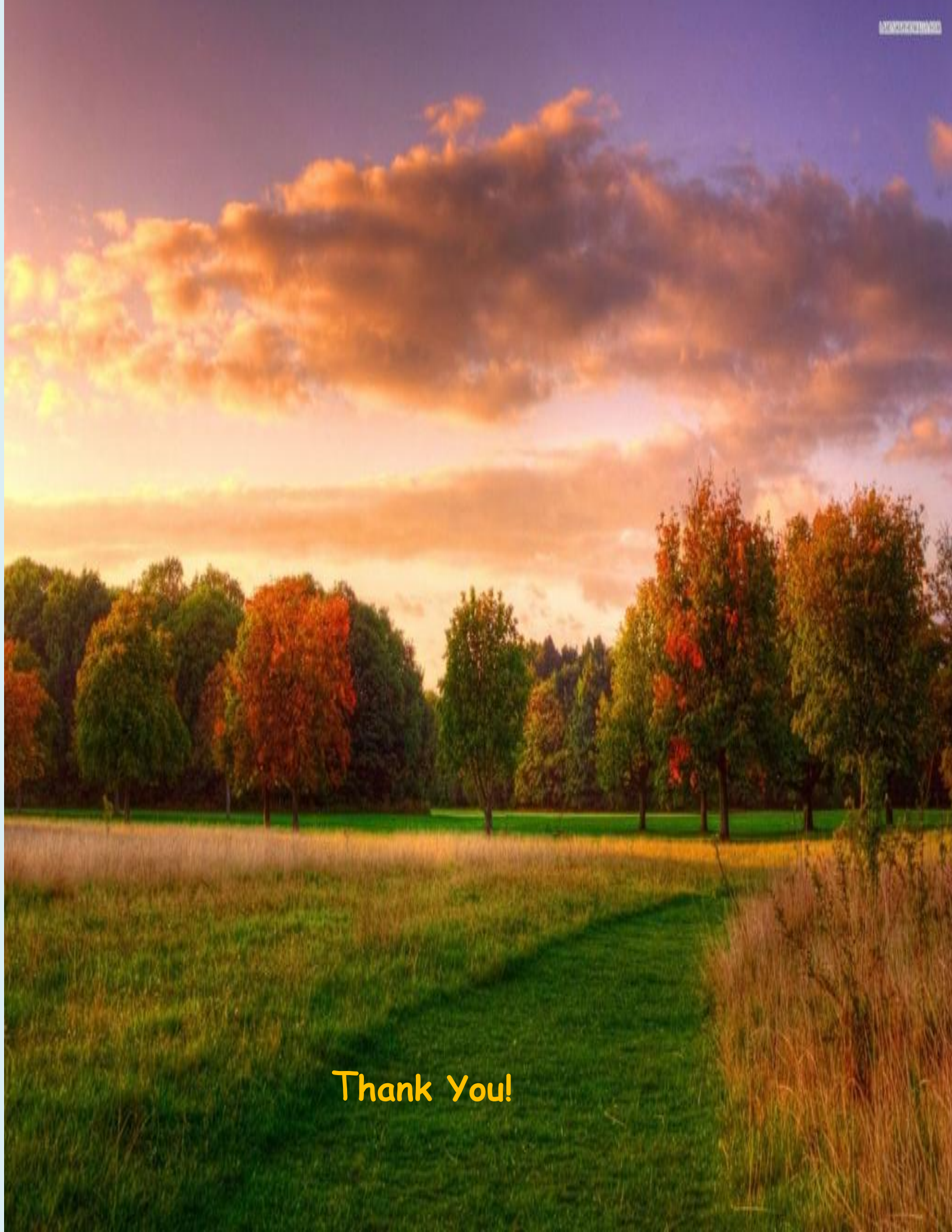
GLUT11: has been cloned based on genomic information and is expressed in three isoforms. Its physiological role in humans has not been identified.

GLUT12: has been identified in the skeletal muscle, heart, small intestine, and prostate. The amount of GLUT12 found in the cellular membrane is increased by insulin (similar to GLUT4) in the normal individual, but not under conditions of obesity and type 2 diabetes that cause resistant insulin receptors. It does not appear to be stored in storage vesicles and does not undergo cycling between storage vesicles and the membrane like GLUT4 does. The affinity of GLUT12 for glucose is unknown at this time, but it does transfer glucose, so there is some degree of affinity.

GLUT13: is highly expressed in certain regions of the brain and to a lower extent in adipose tissue and the kidney. This GLUT has no carbohydrate transport activity. It appears to transport hydrogen ions (protons; H¹) and inositol. Its physiological role is not completely understood at this time.

GLUT14: is similar (if not identical) to GLUT3 and possibly serves as a backup for that GLUT. It is expressed in the testis.

GLUT1	• Blood • Blood-Brain Barrier • Heart (lesser extent) 	• Insulin-Independent
GLUT2	• Liver • Pancreas • Small Intestine 	• Insulin-Independent • High K_m • Low Affinity
GLUT3	• Brain • Neurons • Sperm 	• Insulin-Independent • Low K_m • High Affinity
GLUT4	• Skeletal Muscle • Adipose Tissue • Heart 	

A full-page landscape photograph capturing a serene autumn scene. In the foreground, a lush green field is partially covered with tall, golden-brown grasses. A soft shadow of a person is cast across the grass. The middle ground features a line of trees with foliage in various shades of green, orange, and red. The background is dominated by a vast, dramatic sky filled with large, textured clouds illuminated by the warm, golden light of a setting or rising sun. The overall mood is peaceful and contemplative.

Thank You!